Elevated Salience Network Connectivity is Associated with Increasing Emotional Empathy in Amyloid-β Positive, Healthy Older Adults

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Summary Amyloid-β (Aβ) aggregation is a primary neuropathological feature of Alzheimer's disease (AD). Healthy older adults who had a positive amyloid-β (Aβ+) PET scan, an AD biomarker, showed greater longitudinal increases in emotional empathy than healthy older adults with a negative amyloid-β (Aβ-) scan. In Aβ+ participants, increasing emotional empathy related to elevated connectivity in the salience network (SN), a brain network supporting emotion and empathy.

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Introduction

Abnormal Aβ aggregation in the brain is a hallmark feature of AD that begins decades before cognitive symptoms emerge. Healthy older adults with elevated Aβ—often considered a biomarker of preclinical disease—can be identified during life with molecular PET imaging. Our prior work has shown that emotional contagion, a self-oriented form of emotional empathy, is enhanced in AD and mild cognitive impairment (Sturm et al., 2013). In AD, despite decline in neural systems that support cognitive processes (e.g., episodic memory), there is heightened connectivity in the SN, a system that supports emotion generation and sensation as well as emotional empathy (Zhou et al., 2010). There is some evidence for early SN enhancement in Aβ+ healthy older adults (Schultz et al., 2017), but whether this relates to emotional empathy is unknown.

Aims

We investigated whether Aβ+ healthy older adults had greater longitudinal increases in emotional empathy than Aβ- healthy older adults. We hypothesized that gains in emotional empathy in Aβ+ participants would be related to greater SN connectivity.

Methods

Participants were 86 cognitively normal older adult volunteers followed in a longitudinal study of healthy aging. Informants rated participants' current emotional empathy using the Interpersonal Reactivity Index (Davis, 1983). Emotional empathy is an autonomic form of affect sharing and was measured with two subscales: empathic concern, an other-oriented variant, and personal distress, a self-oriented variant. ^{18}F-AV-45 (Florbetapir) and ^{11}C-Pittsburgh compound B (^{11}C-PiB) PET scans were used to determine that our sample included 23 Aβ+ and 63 Aβ- participants. Functional connectivity matrices were calculated from resting state fMRI data and connectivity between SN node-pairs was related to empathic concern trajectories.

Results

Mixed-model regression analyses revealed that Aβ+ and Aβ- participants did not differ in baseline empathic concern (p = .086) or personal distress (p = .669). Longitudinal analyses found that Aβ+ participants demonstrated greater increases in empathic concern over time than Aβ- participants (p = .043; Fig. 1A). There were no longitudinal differences in personal distress (p = .934). SN connectivity was elevated in Aβ+ participants (p = .016). Within the SN, connectivity between anterior cingulate gyrus (ACC) and periaqueductal gray (PAG)—regions critical for emotion generation—predicted increasing empathic concern over time (left ACC-PAG p = .019; right ACC-PAG p = .004; Fig. 1B).

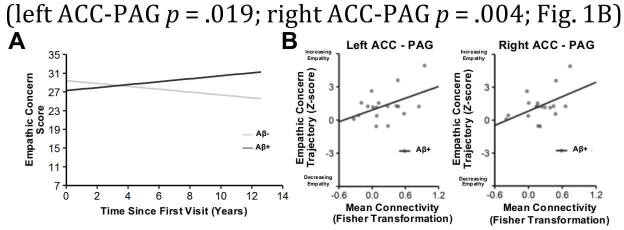

Figure 1. A) Empathic concern across time and B) in relation to SN connectivity.

Conclusions

Aβ+ participants showed greater longitudinal increases in empathic concern than Aβ- participants, and these emotional empathy gains were related to greater SN connectivity. These findings suggest emotional empathy increases in the very early stages of AD and is accompanied by heightened connectivity in SN nodes important for emotion generation.

References

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